

LS INVENTORY

Installation, Configuration & Operations Guide

Multi-Cabinet Architecture - Central Server - Device Pairing - Licensing - Maintenance

Version	1.0 - 29 August 2026
Primary Server	Raspberry Pi 1 + HMI Lab Cabinet
Primary Device	LS_Cab_LabHMI
Central API	Port 3100

This document covers the primary Raspberry Pi and additional cabinet installation, RC522/camera/TFT configuration, Tailscale, Web Admin, device pairing, Trial/Lifetime licensing, offline synchronization, backup, and troubleshooting.

1. System Overview

LS Inventory is a Raspberry Pi based multi-cabinet inventory system. Raspberry Pi 1 currently acts as both the primary Central server and the HMI Lab cabinet. Additional Raspberry Pis operate as cabinet clients and communicate with Central through the private API/Tailscale.

Component	Role
Raspberry Pi 1	Central Server + HMI Lab Cabinet
Central :3100	Web Admin, Central API, central.db
Cabinet :3000	Kiosk, NFC, QR, Borrow/Return/Consume
device.db	Local database on every cabinet
Tailscale	Private device-to-device network

Architecture

WEB ADMIN	PRIMARY PI	LICENSE SERVER
Admin browser	Central :3100 central.db HMI Cabinet :3000	logisourcedigital.web.id
CABINET 2 / 3 / ...	TAILSCALE	LOCAL OPERATION

Important principle: central.db exists only on the primary server. Every cabinet owns its own device.db so it can continue operating during a temporary network/Central outage.

2. Hardware Requirements

- Raspberry Pi with 40-pin GPIO, CSI camera support, and stable networking.
- RC522 13.56 MHz NFC reader on SPI1.
- CSI camera (OV5647 or a camera compatible with rpical).
- 3.5-inch ILI9486/tft35a TFT on SPI0 (matching the current cabinet).
- At least a 32 GB high-endurance/Class 10 microSD; 64 GB is preferred.
- Quality power supply appropriate for the Raspberry Pi model.
- Tailscale for private connectivity between cabinets and Central.

Recommended Raspberry Pi Models

Model	Rating	Reason
Raspberry Pi 4 Model B 2GB/4GB	RECOMMENDED	Best balance: more RAM, USB 3, Gigabit Ethernet, mature CSI/GPIO stack.
Raspberry Pi 5 2GB/4GB	RECOMMENDED / FUTURE	Fastest option; requires good power/cooling. Validate existing TFT overlay/camera cable compatibility.
Raspberry Pi 3 Model B+ 1GB	MINIMUM SAFE	Compatible architecture and 40-pin GPIO; slower but acceptable for a cabinet client.
Raspberry Pi 3 Model B 1GB	WORKS	Current primary hardware proves it can run; not the preferred choice for a new purchase.

Raspberry Pi 3 Model A+ 512MB	NOT RECOMMENDED	No Ethernet, one USB port, only 512MB RAM.
Raspberry Pi Zero 2 W 512MB	NOT RECOMMENDED	512MB RAM, no Ethernet, one USB OTG; Chromium + camera + Node workload is tight.
Raspberry Pi 2 Model B	NOT RECOMMENDED	Older platform; revisions 1.1/1.2 reached EoL in 2025, no onboard Wi-Fi, limited headroom.

Purchase recommendation: use Raspberry Pi 4 Model B 2GB or 4GB for cabinet #2/#3 as the best balance. Raspberry Pi 5 is suitable for future-proofing, but needs better power/cooling and display/camera connector compatibility checks. Models below Raspberry Pi 3B are not recommended for a production LS Inventory build.

3. Cabinet Hardware Wiring

RC522 - SPI1

RC522	Raspberry Pi
3.3V	Pin 1 / 3.3V
GND	Pin 6 / GND
SCK	GPIO21 / Pin 40 / SPI1 SCLK
MOSI	GPIO20 / Pin 38 / SPI1 MOSI
MISO	GPIO19 / Pin 35 / SPI1 MISO
SDA / SS	GPIO16 / Pin 36 / CS
RST	3.3V
IRQ	Not used

```

dtparam=spi=on
dtoverlay=tft35a:rotate=0
dtoverlay=spil-lcs,cs0_pin=16

```

Do not move RC522 to SPI0 because SPI0 is used by the TFT/touch controller.

4. Primary Raspberry Pi Installation (Pi 1)

Pi 1 runs two PM2 processes: LS_Inventory (Cabinet :3000) and LS_Inventory_Central (Central :3100).

```

sudo apt update
sudo apt install -y git curl python3 python3-spidev python3-opencv chromium xinit

sudo mkdir -p /opt/LS_Inventory
sudo chown -R $USER:$USER /opt/LS_Inventory
git clone https://github.com/fbrhrdynt/LS_Inventory.git /opt/LS_Inventory
cd /opt/LS_Inventory
npm install

```

.env

```
APP_NAME="LS Inventory"
PORT=3000
DEVICE_ID=LS_Cab_LabHMI
DEVICE_NAME="Cabinet Lab HMI"
DEVICE_API_KEY=<DEVICE_KEY>
CENTRAL_API_URL=http://127.0.0.1:3100
CENTRAL_PORT=3100
```

PM2

```
cd /opt/LS_Inventory
pm2 start central/server.js --name LS_Inventory_Central --cwd /opt/LS_Inventory
pm2 start app.js --name LS_Inventory --cwd /opt/LS_Inventory
pm2 save
curl http://127.0.0.1:3100/health
```

5. Installing an Additional Cabinet (Pi 2, Pi 3, ...)

A new cabinet clones the same repository, but does NOT run central/server.js and does NOT copy central.db/device.db/.env from another Pi.

```
sudo mkdir -p /opt/LS_Inventory
sudo chown -R $USER:$USER /opt/LS_Inventory
git clone https://github.com/fbrhrdynt/LS_Inventory.git /opt/LS_Inventory
cd /opt/LS_Inventory
npm install
```

Tailscale

```
tailscale status
curl http://<PRIMARY_PI_TAILSCALE_IP>:3100/health
```

Pairing

1. Open Web Admin > Devices & Maintenance > Add Device.
2. Create a unique DEVICE_ID, e.g. LS_Cab_Workshop01.
3. Copy the 6-digit single-use pairing code.
4. On the new Pi run node scripts/pair_device.js.
5. Enter the Central URL/Tailscale address and pairing code.
6. The Pi stores its own DEVICE_API_KEY and performs initial sync.

```
cd /opt/LS_Inventory
node scripts/pair_device.js
node services/google_sync.js
pm2 start app.js --name LS_Inventory --cwd /opt/LS_Inventory
pm2 save
```

Note: services/google_sync.js keeps its legacy filename for compatibility, but it now performs Central Sync, not Google Sheets synchronization.

6. Web Admin and Roles

Menu	Administrator	Admin Staff
Dashboard	Yes	Yes
Inventory	Yes	Yes
Users	Yes	Yes
Transactions	Yes	Yes

Devices & Maintenance	Yes	No
Admin Users	Yes	No
Settings	Yes	No

Administrators can create Web Admin accounts with Administrator or Admin Staff roles. Admin Staff access is restricted in both the UI and API.

Inventory delete

Item deletion is a soft-delete. Transaction history is preserved. A BORROWED item must be returned before it can be deleted.

7. Licensing - Trial and Lifetime

Each Raspberry Pi cabinet has its own hardware fingerprint and license. A new Raspberry Pi automatically starts in TRIAL when no Lifetime license is configured. The current build defaults to a 14-day trial.

Plan	Behavior
TRIAL	Full operational access while active; expires according to configured trial days.
LIFETIME	Persistent full access after successful activation/verification.

Licenses are managed from Devices & Maintenance > License Details, and only Administrators may enter a license. For remote Pi 2/3 devices, the activation command is delivered through Central and processed by the target device.

License endpoints: `logisourcedigital.web.id/api/public/license/verify` and `/activate`. Never commit license keys to GitHub.

8. Cabinet Operations

- NFC login through RC522.
- QR scans use the item UUID.
- Manual CODE uses ITEM_00000 through ITEM_99999 (5 digits).
- BORROWABLE: Borrow -> BORROWED; Return -> AVAILABLE.
- CONSUMABLE: quantity reduces stock; LOW_STOCK/OUT_OF_STOCK follow the configured threshold.
- Pending transactions stay local while Central is offline.

9. Synchronization and Offline Mode

Master data flows incrementally from Central to device.db. device.db then exports JSON caches for fast lookups. If Central is unavailable, the cabinet continues using local data and its transaction queue.

```
cd /opt/LS_Inventory
node services/google_sync.js

# health
curl http://127.0.0.1:3000/health
curl http://<CENTRAL_HOST>:3100/health
```

10. Backup & Restore

Back up databases before upgrades. Databases and .env must never be committed to GitHub.

```
mkdir -p /opt/LS_Inventory/backups
cp /opt/LS_Inventory/central/data/central.db /opt/LS_Inventory/backups/central-$(date +%Y%m%d-%H%M%S).db
cp /opt/LS_Inventory/data/device.db /opt/LS_Inventory/backups/device-$(date +%Y%m%d-%H%M%S).db
```

11. Storage & Camera Maintenance

The primary Pi previously accumulated many rpicas-vid core dumps (~287 MB each), pushing disk usage above 90%. Once identified, these dumps can be removed and core generation should be disabled on a production appliance.

```
cd /opt/LS_Inventory
find . -maxdepth 1 -type f -name 'core.*' -print -delete
sync
df -h /

sudo tee /etc/security/limits.d/99-lsinventory-nocore.conf >/dev/null <<'EOF'
logisource soft core 0
logisource hard core 0
EOF
```

Recommended disk thresholds

Disk Usage	Status
< 75%	NORMAL
75-84%	WARNING
85-89%	HIGH
>= 90%	CRITICAL

12. Safe GitHub Workflow

Repository: github.com/fbrhrdynt/LS_Inventory. Always inspect staged files before pushing.

```
cd /opt/LS_Inventory
git status --short
git diff --cached --name-only

# NEVER stage:
# .env
# data/device.db
# central/data/central.db
# data/license.json
# core.*

git add README.md docs assets scripts services central app.js views public package.json package-lock.json
git diff --cached --name-only
git commit -m "Update LS Inventory documentation and production assets"
git push origin main
```

.gitignore minimum

```
.env
node_modules/
data/*
!data/.gitkeep
central/data/*
!central/data/.gitkeep
cache/*.json
*.db
*.db-wal
*.db-shm
core.*
backups/
*.tar.gz
*.zip
*.log
```

13. New Cabinet Checklist

- ☐ Unique DEVICE_ID
- ☐ Unique device credential / pairing complete
- ☐ Tailscale connected
- ☐ Central :3100 reachable
- ☐ SPI1 /dev/spidev1.0 exists
- ☐ RC522 reports Version 0x92
- ☐ Camera works
- ☐ TFT works
- ☐ Initial Central sync complete
- ☐ NFC login works
- ☐ QR UUID works
- ☐ 5-digit CODE works
- ☐ Borrow/Return/Consume tested
- ☐ License status visible
- ☐ PM2 saved for boot

14. Official Hardware References

- Raspberry Pi 4 Model B specifications - raspberrypi.com/products/raspberry-pi-4-model-b/specifications/
- Raspberry Pi 5 specifications - raspberrypi.com/products/raspberry-pi-5/
- Raspberry Pi 3 Model B+ - raspberrypi.com/products/raspberry-pi-3-model-b-plus/
- Raspberry Pi computer hardware documentation - raspberrypi.com/documentation/computers/raspberry-pi.html
- Raspberry Pi Zero 2 W - raspberrypi.com/products/raspberry-pi-zero-2-w/

Compatibility note: Raspberry Pi 5 uses different MIPI camera/display connectors from Pi 3/4, so the current camera cable/adaptor and TFT overlay should be validated before deployment.

15. Branding Information

LogiSource Digital

Digital. Reliable. Connected

The HD logo is available at assets/LogiSource_Digital_Logo_HD.png and can be displayed in the GitHub README. An LS Inventory branding concept is also included as a visual draft, not as a replacement for the primary brand identity without approval.